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Swing-Up of a Spherical Pendulum on a 7-Axis Industrial Robot

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Abstract: This work demonstrates the swing-up of a custom-built spherical pendulum, which is mounted on the 7-axis industrial robot Kuka LWR IV+. The swing-up trajectory for this system with nine degrees of freedom is calculated offline in a three-step process: First, an optimal control problem for the pendulum model with fewer states and inputs is solved. Then, using the inverse kinematics of the robot model and the forward dynamics of the complete system, this trajectory is converted to a feasible starting solution for the third step, which is solving the optimal control problem for the complete system. Finally, the swing-up is shown experimentally using a two-degrees-of-freedom control structure comprising the calculated trajectory for the feedforward control and a time-variant linear quadratic regulator (LQR) as a feedback controller.

Keywords: Spherical Inverted Pendulum, Redundant Robot, Optimal Control Problem, Optimization, Linear Quadratic Regulator.

1. INTRODUCTION

A quick search on "Inverted Pendulum" in the literature reveals over 50 000 publications. Still, this kind of system remains popular, as new approaches, ideas and systems have been published recently, see, e. g., Hamza et al. (2019). An inverted pendulum comprises many system-theoretic properties like nonlinearity, unstable equilibria, nonminimum-phase behavior and underactuation, which promote it to a benchmark system for advanced control concepts.

In the 1970s, research interest was not only to stabilize a pendulum in its unstable equilibrium, but to also perform the swing-up, as for instance stated in Lundberg and Barton (2010). The first publication on this aspect, Mori et al. (1976), solved an optimal control problem by finding the optimal switching points for a bang-bang input. The swing-up of a pendulum with the most links in experiment has been demonstrated by Glück et al. (2013) with a triple-pendulum, which uses the output-constrained feedforward control design proposed by Graichen and Zeitz (2008). While the swing-up problem of the latter is hard to solve, it relies on a custom-designed actuator, which is able to perform the dynamic motion. In contrast, using an industrial robot to swing up a pendulum imposes many additional kinematic and dynamic constraints, like joint position, velocity and torque limits. Nevertheless, the swing-up of a single-axis pendulum on an industry robot was demonstrated by Winkler and Suchý (2009) using a simple, intuitive control scheme.

Stabilization of a (two-axis) spherical pendulum, without swing-up, was shown by Schreiber et al. (2001) on the experimental robot "DLR LWR II". The custom-built highly dynamic "juggling robot $2k\pi$" was used in Lenoir et al. (1998) to swing up a spherical pendulum using flatness-based trajectory planning and in Albouy and Praly (2000) using dynamic invariants and forwarding. The simulation studies contributed by Shiriaev et al. (1999, 2004) rely on modifying the system equilibrium properties in the closed-loop system by a nonlinear switching feedback law. However, the swing-up of a spherical pendulum on an industrial robot has not been demonstrated yet, to the best of the author's knowledge. Additionally, no previous work could be revealed, which calculates a swing-up trajectory for the spherical pendulum using an optimal control problem.

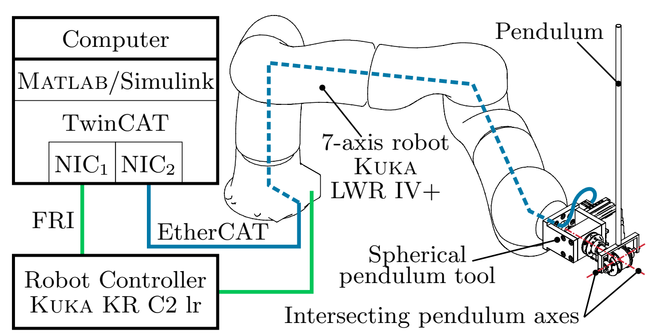

Fig. 1. System overview of the experimental setup.

The remainder of this work is organized as follows: In Section 2, the experimental setup and the mechatronic design of the spherical pendulum are described in detail. Section 3 derives two separate rigid-body models for the robot and the pendulum, which are then combined to constitute the complete system model. The main contri-

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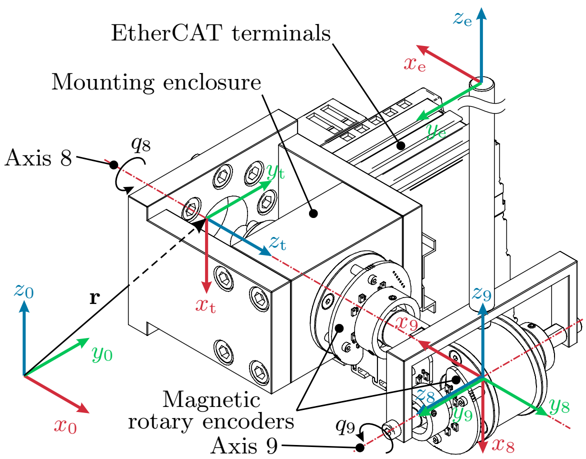

Fig. 2. Custom-built spherical pendulum in the unstable equilibrium with coordinate frames.

bution of this work, the optimization-based calculation of the swing-up trajectory, is presented as a three-step process in Section 4. Using the linear quadratic regulator (LQR) designed in Section 5, the swing-up is shown experimentally in Section 6. A video of the swing-up and stabilization of the spherical pendulum can be found at `www.acin.tuwien.ac.at/9e6f`. Finally, Section 7 concludes this paper and proposes ideas for future works.

2. EXPERIMENTAL SETUP AND PENDULUM: MECHATRONIC DESIGN

The experimental setup is composed of four components, as depicted in Fig. 1: the custom-built spherical pendulum tool, the 7-axis robot KUKA LWR IV+, the robot controller KUKA KR C2 lr, and the computer running MATLAB/Simulink modules via the real-time automation software BECKHOFF TwinCAT. The computer is equipped with two network interface cards (NIC), which communicate with the robot controller using the Fast Research Interface (FRI) and with the pendulum sensors via EtherCAT. The FRI control strategy "Joint Specific Impedance Control" is used to control the robot, whereas the stiffness and damping parameters are set to zero to obtain a torque interface to the robot, see KUKA Roboter GmbH (2010). The experimental setup uses a sampling time of $T_s = 1\,\mathrm{ms}$ for the controller and the two network interfaces FRI and EtherCAT.

The custom-built spherical pendulum comprises a mounting enclosure and two intersecting axes (i.e. Axis 8 and Axis 9 in Fig. 2), which rotate freely using single row ball bearings to minimize friction. Both axes are equipped with 14 bit magnetic rotary encoders (RLS ORBIS™ absolute rotary encoder) to obtain relative angle measurements between the magnetic actuators and the respective read head. While the sensor for Axis 8 is connected directly to the EtherCAT terminals, the wires of the Axis 9 sensor run through a boring along Axis 8 and a slip ring, which allows for an infinite travel in Axis 8. The sensors are interfaced by the terminal BECKHOFF EL5002 via the synchronous serial interface (SSI), which communicates with the computer via EtherCAT using the coupler BECKHOFF EK1100.

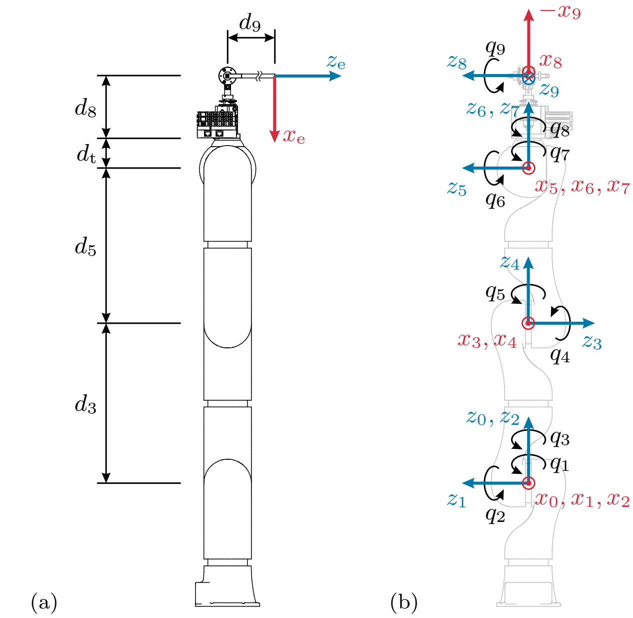

Fig. 3. Schematic drawing of the mechanical system and its corresponding coordinate frames. (a) Side view. (b) Front view.

Table 1. Denavit-Hartenberg parameters of the complete system.

i	a_i	α_i / rad	d_i / m	θ_i
1	0	$\pi/2$	0	q_1
2	0	$-\pi/2$	0	q_2
3	0	$-\pi/2$	0.4	q_3
4	0	$\pi/2$	0	q_4
5	0	$\pi/2$	0.39	q_5
6	0	$-\pi/2$	0	q_6
7	0	0	0	q_7
t	0	0	0.078	0
8	0	$\pi/2$	0.157	q_8
9	0	$\pi/2$	0	$q_9 - \pi/2$
e	0	0	0.535	0

3. MATHEMATICAL MODELS

In the course of this section, three rigid-body models are derived, i. e. the complete mechanical system, which comprises the robot (with subscript "r") and the pendulum (with subscript "p"). The coordinate frames of all components are shown in Figs. 2 and 3, and the kinematics are described by the corresponding DH parameters listed in Tab. 1.

The robot is modeled as a rigid-body system in the state space using the seven generalized coordinates $\mathbf{q}_\mathrm{r}$ of the robot as

$$\begin{aligned}\dot{\mathbf{x}}_\mathrm{r} &= \mathbf{f}_\mathrm{r}(\mathbf{x}_\mathrm{r}, \mathbf{u}_\mathrm{r}) \\ &= \begin{bmatrix} \dot{\mathbf{q}}_\mathrm{r} \\ \mathbf{D}_\mathrm{r}^{-1}(\mathbf{q}_\mathrm{r})(-\mathbf{C}_\mathrm{r}(\mathbf{q}_\mathrm{r}, \dot{\mathbf{q}}_\mathrm{r})\dot{\mathbf{q}}_\mathrm{r} - \mathbf{g}_\mathrm{r}(\mathbf{q}_\mathrm{r}) + \mathbf{u}_\mathrm{r}) \end{bmatrix},\end{aligned} \tag{1}$$

with the system state $\mathbf{x}_\mathrm{r}^\mathrm{T} = \left[\mathbf{q}_\mathrm{r}^\mathrm{T} \dot{\mathbf{q}}_\mathrm{r}^\mathrm{T}\right] \in \mathbb{R}^{14}$ and the seven torque inputs $\mathbf{u}_\mathrm{r}$. The mass matrix $\mathbf{D}_\mathrm{r}(\mathbf{q}_\mathrm{r})$, the Coriolis matrix $\mathbf{C}_\mathrm{r}(\mathbf{q}_\mathrm{r}, \dot{\mathbf{q}}_\mathrm{r})$, and the gravitational vector $\mathbf{g}_\mathrm{r}(\mathbf{q}_\mathrm{r})$ in (1) derive from the link masses m_i, inertia tensors $\mathbf{I}_i$ and the robot kinematics with $i \in \{1, \ldots, 7\}$. The dynamic parameters of the model originate from Gaz et al. (2016). The output function of the robot,

Table 2. Kinematic and dynamic limits for the robot.

Joint i	Angle Limit $\bar{\mathbf{q}}_\mathrm{r}$ / °	Velocity Limit $\bar{\dot{\mathbf{q}}}_\mathrm{r}$ / ° s^{-1}	Torque Limit $\bar{\mathbf{u}}_\mathrm{r}$ / N m
1	170	112.5	200
2	120	112.5	200
3	170	112.5	100
4	120	112.5	100
5	170	180	100
6	120	112.5	30
7	170	112.5	30

$$\mathbf{z}_\mathrm{r} = \mathbf{h}_\mathrm{r}(\mathbf{q}_\mathrm{r}) = \begin{bmatrix} \mathbf{h}_{\mathrm{r,t}}(\mathbf{q}_\mathrm{r}) \\ \mathbf{h}_{\mathrm{r,r}}(\mathbf{q}_\mathrm{r}) \end{bmatrix} , \tag{2}$$

is composed by the position $\mathbf{h}_{\mathrm{r,t}}(\mathbf{q}_\mathrm{r})$ and the orientation $\mathbf{h}_{\mathrm{r,r}}(\mathbf{q}_\mathrm{r})$ of the tool frame $(0_\mathrm{t} x_\mathrm{t} y_\mathrm{t} z_\mathrm{t})$. Using the geometric Jacobian $\mathbf{J}_\mathrm{r} = \partial \mathbf{h}_\mathrm{r}(\mathbf{q}_\mathrm{r})/\partial \mathbf{q}_\mathrm{r}$, the time derivatives of $\mathbf{z}_\mathrm{r}$ are calculated as

$$\dot{\mathbf{z}}_\mathrm{r} = \mathbf{J}_\mathrm{r}\dot{\mathbf{q}}_\mathrm{r} , \qquad \ddot{\mathbf{z}}_\mathrm{r} = \dot{\mathbf{J}}_\mathrm{r}\dot{\mathbf{q}}_\mathrm{r} + \mathbf{J}_\mathrm{r}\ddot{\mathbf{q}}_\mathrm{r} , \tag{3}$$

which are used for the inverse kinematics in Section 4. Finally, the kinematic and dynamic limits of the robot, as documented in Kuka Roboter GmbH (2010), are listed in Tab. 2.

Next, a separate model of the spherical pendulum is derived with the coordinate frames in Fig. 2, i. e. the rows $i \in \{\mathrm{t}, 8, 9, \mathrm{e}\}$ in Tab. 1. This model, denoted by subscript “p”, is controlled by the Cartesian acceleration $\ddot{\mathbf{r}}$ of the origin of the tool frame $(0_\mathrm{t} x_\mathrm{t} y_\mathrm{t} z_\mathrm{t})$, while the orientation of the tool frame stays fixed, thus the input is $\mathbf{u}_\mathrm{p} = \ddot{\mathbf{r}}$. The vector of generalized coordinates $\mathbf{q}_\mathrm{p}^\mathrm{T} = [q_8 \;\, q_9]$ contains the coordinates of the pendulum axes. Hence, the resulting model with the state $\mathbf{x}_\mathrm{p}^\mathrm{T} = \left[\mathbf{q}_\mathrm{p}^\mathrm{T}\dot{\mathbf{q}}_\mathrm{p}^\mathrm{T}\mathbf{r}^\mathrm{T}\dot{\mathbf{r}}^\mathrm{T}\right] \in \mathbb{R}^{10}$ takes the form

$$\dot{\mathbf{x}}_\mathrm{p} = \mathbf{f}_\mathrm{p}(\mathbf{x}_\mathrm{p}, \mathbf{u}_\mathrm{p}) , \tag{4}$$

which is derived using the Euler-Lagrange formalism. The input $\mathbf{u}_\mathrm{p}$, i. e. the Cartesian acceleration $\ddot{\mathbf{r}}$, induces forces on the pendulum which are systematically considered.

The complete mechanical system (without subscript), comprising the robot and the pendulum reads as

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}) = \begin{bmatrix} \dot{\mathbf{q}} \\ \mathbf{D}^{-1}(\mathbf{q})(-\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{g}(\mathbf{q}) + \boldsymbol{\tau}) \end{bmatrix} , \tag{5}$$

with $\mathbf{x}^\mathrm{T} = \left[\mathbf{q}^\mathrm{T} \;\, \dot{\mathbf{q}}^\mathrm{T}\right] \in \mathbb{R}^{18}$ and $\mathbf{q}^\mathrm{T} = \left[\mathbf{q}_\mathrm{r}^\mathrm{T} \;\, \mathbf{q}_\mathrm{p}^\mathrm{T}\right]$. The seven torque inputs of the robot $\mathbf{u}$ are applied to the system as generalized forces $\boldsymbol{\tau}$

$$\boldsymbol{\tau}^\mathrm{T} = \left[\mathbf{u}^\mathrm{T} \;\, 0 \;\, 0\right] . \tag{6}$$

The terms $\mathbf{D}(\mathbf{q})$, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$, and $\mathbf{g}(\mathbf{q})$ denote the mass matrix, the Coriolis matrix and the gravitational vector of the complete model, respectively. Note that this model incorporates all the couplings between the rigid bodies of the robot and the pendulum. Moreover, for the model (5), the mass and inertia of the pendulum mounting enclosure are included in the link 7, and thus, the coordinate frame “t” in Tab. 1 incorporates no inertia. The kinematic and dynamic limits for the robot in Tab. 2 are complemented by two further entries for the pendulum axes as

$$\bar{\mathbf{q}}^\mathrm{T} = \left[\bar{\mathbf{q}}_\mathrm{r}^\mathrm{T} \;\, 360° \;\, 360°\right] \tag{7a}$$

$$\bar{\dot{\mathbf{q}}}^\mathrm{T} = \left[\bar{\dot{\mathbf{q}}}_\mathrm{r}^\mathrm{T} \;\, 720\,°\,\mathrm{s}^{-1} \;\, 720\,°\,\mathrm{s}^{-1}\right] \tag{7b}$$

$$\bar{\mathbf{u}} = \bar{\mathbf{u}}_\mathrm{r} . \tag{7c}$$

Additionally, these vectors are also used for scaling in the plots. The pendulum axes actually allow infinite travel and the corresponding values in (7) are set generously to not impose any limits. The friction is neglected in all three models.

4. SWING-UP TRAJECTORY

The aim of this section is to find a trajectory, which transitions the complete system (5) from an initial robot pose $\mathbf{q}_{\mathrm{r},0}$ and the hanging pendulum $\mathbf{q}_{\mathrm{p},0}^\mathrm{T} = [\pi \;\, 0]$ at $t_0 = 0$,

$$\mathbf{q}_0^\mathrm{T} = \left[\mathbf{q}_{\mathrm{r},0}^\mathrm{T} \;\, \mathbf{q}_{\mathrm{p},0}^\mathrm{T}\right] = \left[0 \;\, -\tfrac{\pi}{4} \;\, 0 \;\, \tfrac{\pi}{4} \;\, 0 \;\, 0 \;\, 0 \;\, \pi \;\, 0\right] , \tag{8}$$

to a final pose

$$\mathbf{q}_1^\mathrm{T} = \left[\mathbf{q}_{\mathrm{r},1}^\mathrm{T} \;\, \mathbf{q}_{\mathrm{p},1}^\mathrm{T}\right] = \left[\tfrac{\pi}{6} \;\, -\tfrac{\pi}{2} \;\, -\tfrac{\pi}{2} \;\, \tfrac{\pi}{3} \;\, 0 \;\, \tfrac{\pi}{6} \;\, \tfrac{\pi}{2} \;\, 0 \;\, 0\right] \tag{9}$$

with the pendulum in the unstable equilibrium $\mathbf{q}_{\mathrm{p},1} = \mathbf{0}$ at $t_1 = 2.5\,\mathrm{s}$. For both $\mathbf{q}_{\mathrm{r},0}$ and $\mathbf{q}_{\mathrm{r},1}$ to be valid, the tool frame orientation has to match Fig. 2. The poses for the robot are chosen such that collisions with the surroundings are avoided with the hanging pendulum and that the final pose provides high manipulability to balance the pendulum.

Due to the high dimensionality of the states and the inputs of the system, it is not easy to find a swing-up trajectory without a good initial guess. Hence, this work resorts to a three-step process: First, an optimal control problem for the pendulum model (4), which constitutes a reduced number of states and inputs, is solved. Second, inverse kinematics of the robot model (1) and forward dynamics of the complete model (5) are used to convert the trajectory into the joint space of the robot. This results in a viable starting solution for the final step, i. e. the optimal control problem for the complete model (5).

4.1 Simplified Optimal Control Problem

The optimal swing-up control $\mathbf{u}_\mathrm{p}(t)$ for the spherical pendulum minimizes the cost functional $J(\mathbf{u}_\mathrm{p}(\cdot))$ with respect the system equations (4) with a given initial condition, i. e.

$$\min_{\mathbf{u}_\mathrm{p}(\cdot)} \quad J(\mathbf{u}_\mathrm{p}(\cdot)) = \int_{t_0}^{t_1} l_\mathrm{p}(t, \mathbf{x}_\mathrm{p}(t), \mathbf{u}_\mathrm{p}(t))\,\mathrm{d}t \tag{10a}$$

$$\text{s.t.} \quad \dot{\mathbf{x}}_\mathrm{p} = \mathbf{f}_\mathrm{p}(\mathbf{x}_\mathrm{p}, \mathbf{u}_\mathrm{p}) , \qquad \mathbf{x}_\mathrm{p}(t_0) = \mathbf{x}_{\mathrm{p},0} \tag{10b}$$

$$\boldsymbol{\phi}_\mathrm{p}(t_1, \mathbf{x}_\mathrm{p}(t_1)) = \mathbf{0} , \tag{10c}$$

where $l_\mathrm{p}(\cdot)$ is the Lagrange function. Further constraints are the terminal conditions $\boldsymbol{\phi}_\mathrm{p}(\cdot)$. Using the direct simultaneous method, the states $\mathbf{x}_\mathrm{p}^k = \mathbf{x}_\mathrm{p}(t^k)$ and the input $\mathbf{u}_\mathrm{p}^k = \mathbf{u}_\mathrm{p}(t^k)$ are discretized on the equidistant time grid with N grid points

$$t^k = t_0 + (k-1)\frac{t_1 - t_0}{N-1}, \quad k = 1, \ldots, N . \tag{11}$$

Thus, the infinite-dimensional dynamic optimization problem is transformed to a finite-dimensional static problem

$$\min_{\mathbf{y}_\mathrm{p}} \quad J_\mathrm{p}(\mathbf{y}_\mathrm{p}) \tag{12a}$$

$$\text{s.t.} \quad \mathbf{c}_\mathrm{p}(\mathbf{y}_\mathrm{p}) = \mathbf{0} , \tag{12b}$$

with the optimization variable $\mathbf{y}_\mathrm{p} \in \mathbb{R}^{(10+3)N}$

$$\mathbf{y}_\mathrm{p}^\mathrm{T} = \left[\left(\mathbf{x}_\mathrm{p}^1\right)^\mathrm{T} \;\, \left(\mathbf{u}_\mathrm{p}^1\right)^\mathrm{T} \;\, \cdots \;\, \left(\mathbf{x}_\mathrm{p}^N\right)^\mathrm{T} \;\, \left(\mathbf{u}_\mathrm{p}^N\right)^\mathrm{T}\right] , \tag{13}$$

whereas (10b) and (10c) are discretized yielding the equality constraints (12b). In the following, the superscript is used to refer to the corresponding time step. The cost functional (10a) and the system equations (10b) are discretized using the trapezoidal rule, resulting in

$$J(\mathbf{y}_\mathrm{p}) = \sum_{k=1}^{N-1} \frac{1}{2}\left(t^{k+1} - t^k\right)\Big(l_\mathrm{p}\left(t^k, \mathbf{x}_\mathrm{p}^k, \mathbf{u}_\mathrm{p}^k\right) + l_\mathrm{p}\left(t^{k+1}, \mathbf{x}_\mathrm{p}^{k+1}, \mathbf{u}_\mathrm{p}^{k+1}\right)\Big) \tag{14}$$

with quadratic cost

$$l_\mathrm{p}\left(t^k, \mathbf{x}_\mathrm{p}^k, \mathbf{u}_\mathrm{p}^k\right) = \frac{1}{2}\Big(\left(\mathbf{x}_\mathrm{p}^k - \tilde{\mathbf{x}}_\mathrm{p}\right)^\mathrm{T}\mathbf{Q}_\mathrm{p}\left(\mathbf{x}_\mathrm{p}^k - \tilde{\mathbf{x}}_\mathrm{p}\right) + \left(\mathbf{u}_\mathrm{p}^k - \tilde{\mathbf{u}}_\mathrm{p}\right)^\mathrm{T}\mathbf{R}_\mathrm{p}\left(\mathbf{u}_\mathrm{p}^k - \tilde{\mathbf{u}}_\mathrm{p}\right)\Big)\ , \tag{15}$$

and

$$\mathbf{c}_\mathrm{p}(\mathbf{y}_\mathrm{p}) = \begin{bmatrix} \mathbf{x}_\mathrm{p}^2 - \mathbf{x}_\mathrm{p}^1 - \frac{t^2-t^1}{2}\left(\mathbf{f}_\mathrm{p}\left(\mathbf{x}_\mathrm{p}^1, \mathbf{u}_\mathrm{p}^1\right) + \mathbf{f}_\mathrm{p}\left(\mathbf{x}_\mathrm{p}^2, \mathbf{u}_\mathrm{p}^2\right)\right) \\ \vdots \\ \mathbf{x}_\mathrm{p}^N - \mathbf{x}_\mathrm{p}^{N-1} - \frac{t^N-t^{N-1}}{2}\left(\mathbf{f}_\mathrm{p}\left(\mathbf{x}_\mathrm{p}^{N-1}, \mathbf{u}_\mathrm{p}^{N-1}\right) + \mathbf{f}_\mathrm{p}\left(\mathbf{x}_\mathrm{p}^N, \mathbf{u}_\mathrm{p}^N\right)\right) \\ \mathbf{x}_\mathrm{p}^1 - \mathbf{x}_\mathrm{p}(t_0) \\ \mathbf{x}_\mathrm{p}^N - \mathbf{x}_\mathrm{p}(t_1) \end{bmatrix} .$$

In (15), the positive (semi) definite weighting matrices $\mathbf{Q}_\mathrm{p}$ and $\mathbf{R}_\mathrm{p}$ penalize the deviations from the reference values $\tilde{\mathbf{x}}_\mathrm{p}$ and $\tilde{\mathbf{u}}_\mathrm{p}$.

The static optimization problem (12), on the time grid (11) with $N = 80$ grid points, is now used to find the transition from

$$\mathbf{x}_\mathrm{p}(t_0)^\mathrm{T} = \left[\mathbf{q}_{\mathrm{p},0}^\mathrm{T}\ \mathbf{0}\ \mathbf{h}_{\mathrm{r,t}}(\mathbf{q}_{\mathrm{r},0})^\mathrm{T}\ \mathbf{0}\right] \tag{16}$$

to

$$\mathbf{x}_\mathrm{p}(t_1)^\mathrm{T} = \left[\mathbf{q}_{\mathrm{p},1}^\mathrm{T}\ \mathbf{0}\ \mathbf{h}_{\mathrm{r,t}}(\mathbf{q}_{\mathrm{r},1})^\mathrm{T}\ \mathbf{0}\right] . \tag{17}$$

The reference values in (15) are set to $\tilde{\mathbf{x}}_\mathrm{p} = \mathbf{x}_\mathrm{p}(t_1)$ and $\tilde{\mathbf{u}}_\mathrm{p} = \mathbf{0}$ and the weighting matrices are chosen in such a way that the velocities along the z_0 and y_0 axis, as well as the accelerations along the x_0 axis remain small, i. e.

$$\mathbf{Q}_\mathrm{p} = \mathrm{diag}(1, 1, 1, 1, 1, 1, 1, 1, 100, 100) \tag{18a}$$
$$\mathbf{R}_\mathrm{p} = \mathrm{diag}(100, 1, 1) . \tag{18b}$$

4.2 Inverse Kinematics and Forward Dynamics

In the next step, the trajectory from the previous subsection is converted to a starting solution for the complete system (5). The parameters in (18) were chosen such that the kinematic and dynamic limits of the robot (Tab. 2) are not violated.

Inverse Kinematics. The time-discrete trajectory of the tool frame from the previous subsection $(\mathbf{r}^k, \dot{\mathbf{r}}^k, \ddot{\mathbf{r}}^k)$ is now mapped into the robot joint space as $(\mathbf{q}^k, \dot{\mathbf{q}}^k, \ddot{\mathbf{q}}^k)$ for $k = 1, \ldots, N$ by rearranging (3). Note that the Jacobian $\mathbf{J}_\mathrm{r}$ is a 6×7 matrix due to the kinematic redundancy of the robot. The robot trajectory is obtained by using the pseudo inverse, denoted by †,

$$\dot{\mathbf{q}}_\mathrm{r}^k = \left(\mathbf{J}_\mathrm{r}^k\right)^\dagger \begin{bmatrix}\dot{\mathbf{r}}^k \\ \mathbf{0}\end{bmatrix} , \qquad \ddot{\mathbf{q}}_\mathrm{r}^k = \left(\mathbf{J}_\mathrm{r}^k\right)^\dagger \left(\begin{bmatrix}\ddot{\mathbf{r}}^k \\ \mathbf{0}\end{bmatrix} - \dot{\mathbf{J}}_\mathrm{r}^k \dot{\mathbf{q}}_\mathrm{r}^k\right)$$

and applying the Euler method

$$\mathbf{q}_\mathrm{r}^{k+1} = \mathbf{q}_\mathrm{r}^k + \left(t^{k+1} - t^k\right)\dot{\mathbf{q}}_\mathrm{r}^k\ , \quad k = 1, \ldots, N-1\ , \tag{19}$$

with the initial value $\mathbf{q}_\mathrm{r}^1 = \mathbf{q}_{\mathrm{r},0}$.

Forward Dynamics. The calculated trajectory of the robot $(\mathbf{q}_\mathrm{r}^k, \dot{\mathbf{q}}_\mathrm{r}^k, \ddot{\mathbf{q}}_\mathrm{r}^k)$ and the pendulum $(\mathbf{q}_\mathrm{p}^k, \dot{\mathbf{q}}_\mathrm{p}^k)$ are assembled to constitute the complete system trajectory $(\mathbf{q}^k, \dot{\mathbf{q}}^k, \ddot{\mathbf{q}}^k)$ using

$$\ddot{\mathbf{q}}^k = \begin{bmatrix} \ddot{\mathbf{q}}_\mathrm{r}^k \\ [\mathbf{0}_{2\times 2}\ \mathbf{E}_2\ \mathbf{0}_{2\times 6}]\mathbf{f}_\mathrm{p}\left(\mathbf{x}_\mathrm{p}^k, \mathbf{u}_\mathrm{p}^k\right)\end{bmatrix} , \quad k = 1, \ldots, N \tag{20}$$

with the $n\times n$ identity matrix $\mathbf{E}_n$ and the $m\times n$ zero matrix $\mathbf{0}_{m\times n}$. Finally, the dynamic equations for the complete system (5) and (6) are rearranged to find the robot joint torques $\mathbf{u}^k$, which realize the tool motion

$$\boldsymbol{\tau}^k = \mathbf{D}(\mathbf{q}^k)\ddot{\mathbf{q}}^k + \mathbf{C}(\mathbf{q}^k, \dot{\mathbf{q}}^k)\dot{\mathbf{q}}^k + \mathbf{g}(\mathbf{q}^k) \tag{21}$$
$$\mathbf{u}^k = [\mathbf{E}_7\ \mathbf{0}_{7\times 2}]\boldsymbol{\tau}^k\ , \quad k = 1, \ldots, N . \tag{22}$$

4.3 Complete Optimal Control Problem

Analogous to (10), an optimal control problem for the system (5) is formulated and discretized on the time grid (11). The resulting static optimization problem

$$\min_{\mathbf{y}} \quad J(\mathbf{y}) \tag{23a}$$
$$\text{s.t.} \quad \mathbf{c}(\mathbf{y}) = \mathbf{0} \tag{23b}$$
$$\mathbf{y}_\mathrm{lb} \le \mathbf{y} \le \mathbf{y}_\mathrm{ub}\ , \tag{23c}$$

with

$$\mathbf{y}^\mathrm{T} = \left[\left(\mathbf{x}^1\right)^\mathrm{T}\ \left(\mathbf{u}^1\right)^\mathrm{T}\ \cdots\ \left(\mathbf{x}^N\right)^\mathrm{T}\ \left(\mathbf{u}^N\right)^\mathrm{T}\right] \in \mathbb{R}^{(18+7)N} \tag{24}$$

additionally includes box constraints with the lower and upper bound $\mathbf{y}_\mathrm{lb}$ and $\mathbf{y}_\mathrm{ub}$, respectively, to take into account the dynamic and kinematic limits of the robot in Tab. 2. These bounds are composed of the joint position limits $\bar{\mathbf{q}}$, the joint velocity limits $\dot{\bar{\mathbf{q}}}$, and the torque limits $\bar{\mathbf{u}}$ from (7), i. e.

$$\mathbf{y}_\mathrm{ub}^\mathrm{T} = \left[\bar{\mathbf{q}}^\mathrm{T}\ \dot{\bar{\mathbf{q}}}^\mathrm{T}\ \bar{\mathbf{u}}^\mathrm{T}\ \cdots\ \bar{\mathbf{q}}^\mathrm{T}\ \dot{\bar{\mathbf{q}}}^\mathrm{T}\ \bar{\mathbf{u}}^\mathrm{T}\right] , \quad \mathbf{y}_\mathrm{lb} = -\mathbf{y}_\mathrm{ub} .$$

Further, the equilibria (8) and (9) are set for the initial and final condition $\mathbf{x}(t_0)$ and $\mathbf{x}(t_1)$, respectively. Additional kinematic constraints can be included in (23) to prevent collisions of the pendulum tip with the environment. Note that the constraint on the tool frame orientation from Section 4.3 is relaxed in this optimization problem. While $J(\mathbf{y})$ is constructed analogous to (14), the Lagrange function (15) is adapted to

$$l\left(t^k, \mathbf{x}^k, \mathbf{u}^k\right) = \frac{1}{2}\Big(\left(\mathbf{x}^k - \tilde{\mathbf{x}}\right)^\mathrm{T}\mathbf{Q}\left(\mathbf{x}^k - \tilde{\mathbf{x}}\right) + \left(\mathbf{u}^k - \tilde{\mathbf{u}}\right)^\mathrm{T}\mathbf{R}^k\left(\mathbf{u}^k - \tilde{\mathbf{u}}\right)\Big)\ , \tag{25}$$

where the reference values are chosen as $\tilde{\mathbf{x}} = \mathbf{x}(t_1)$ and $\tilde{\mathbf{u}} = \mathbf{g}(\mathbf{q}_1)$. Note that (25) contains a time-variant input weighting matrix $\mathbf{R}^k$, which is chosen as

$$\mathbf{R}^k = \begin{cases}\mathbf{E}_7, & \text{for } t^k \le 2.45\,\mathrm{s} \\ 10^3 \cdot \mathbf{E}_7, & \text{for } t^k > 2.45\,\mathrm{s}\end{cases} \tag{26}$$

to ensure a smooth motion of the complete system into its final position $\mathbf{q}_1$. Moreover, the state weighting matrix $\mathbf{Q}$ is tuned to strongly penalize motions in q_9 and minimize the robot joint velocities such that sufficient reserve remains for all joints to the respective limits, i. e.

$$\mathbf{Q} = \mathrm{diag}(1, 1, 1, 1, 1, 1, 1, 0, 1000, 200, 1, 900, 300, 50, 100, 10, 0, 1000) . \tag{27}$$

In the final step of the swing-up trajectory generation, the optimal control problem (23) is solved by using the

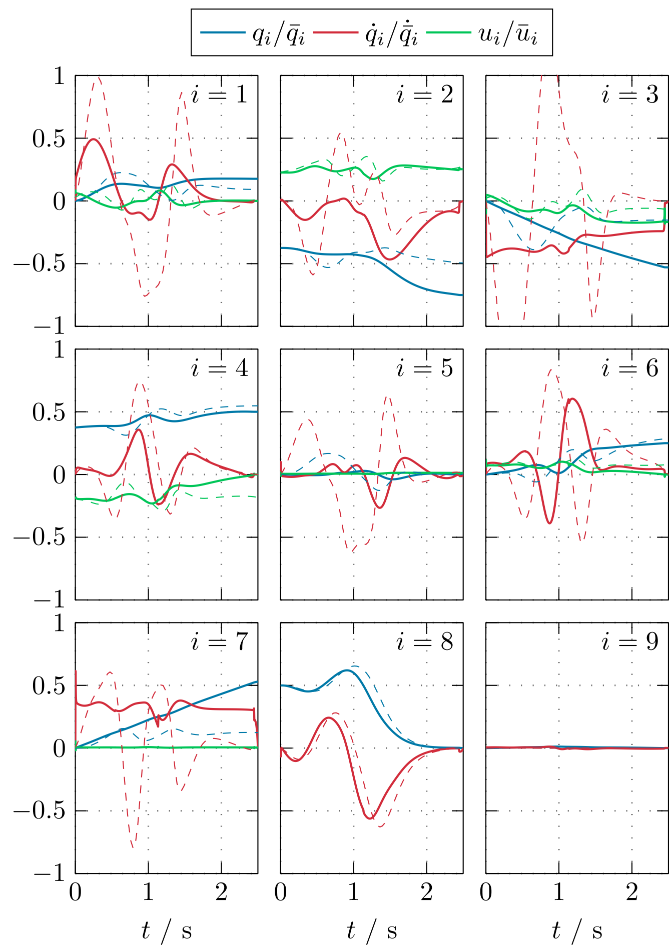

Fig. 4. Swing-up trajectory for the complete optimal control problem. The dashed lines are the scaled initial trajectory from Section 4.2 and the solid lines represent the scaled final solution from Section 4.3 for $N = 2501$.

trajectory from Subsection 4.2 as initial guess in (24). Initially, the optimization starts with $N = 80$, while the number of grid points is successively increased to match the sampling time of the experimental setup ($N = 2501$). In each iteration, N is increased by 10 % and the solution of the previous iteration is interpolated on the new time grid to serve as initial value for the next iteration. Due to memory limitations on the computing hardware, the above iteration could be performed up to $N = 1282$ (32050 optimization variables). Thus, the final optimization result was interpolated to $N = 2501$ using cubic splines.

The resulting trajectory and the initial trajectory from Section 4.2, scaled to the respective limits from (7), are shown in Fig. 4. Both trajectories show a similar pendulum motion (Axes 8 and 9), while the robot motion (Axes 1 to 7) changes drastically. Respecting the robot dynamics, the Cartesian trajectory of Subsection 4.1 is mapped into the joint space, resulting in a suboptimal, infeasible robot motion as initial trajectory for the complete optimal control problem. This final optimization problem refines this motion, relaxes the kinematic constraints, reduces the joint velocities and torques, and increases the reserve to the respective joint limits. Thus, the calculated swing-up trajectory for the complete system (5) is feasible, satisfies all constraints, and provides sufficient control reserves for a closed-loop controller to stabilize the swing-up trajectory.

5. CONTROLLER DESIGN

In this section, a time-variant LQR is designed to stabilize the system (5) around the swing-up trajectory found in Section 4.3. Closed-loop control is required, as unmodeled friction and model uncertainties would prevent a successful pendulum swing-up. Furthermore, the stationary solution of the LQR is used to stabilize the pendulum after the swing-up. Hence, a switching controller is not required.

To apply the LQR to the complete system, the system equations (5) need to be linearized around the desired swing-up trajectory $(\mathbf{x}_\mathrm{d}^k, \mathbf{u}_\mathrm{d}^k)$, subscripted by "d", by introducing the deviations $\mathbf{x}^k = \mathbf{x}_\mathrm{d}^k + \Delta\mathbf{x}^k$ and $\mathbf{u}^k = \mathbf{u}_\mathrm{d}^k + \Delta\mathbf{u}^k$ for the states and input, respectively. The discrete-time system dynamics reads as

$$\Delta\mathbf{x}^{k+1} = \mathbf{\Phi}^k \Delta\mathbf{x}^k + \mathbf{\Gamma}^k \Delta\mathbf{u}^k , \tag{28}$$

with

$$\mathbf{\Phi}^k = \mathbf{E}_{18} + T_\mathrm{s}\mathbf{A}^k , \qquad \mathbf{\Gamma}^k = T_\mathrm{s}\mathbf{B}^k ,$$
$$\mathbf{A}^k = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right|_{\substack{\mathbf{x}=\mathbf{x}_\mathrm{d}^k \\ \mathbf{u}=\mathbf{u}_\mathrm{d}^k}} , \qquad \mathbf{B}^k = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{u}}\right|_{\substack{\mathbf{x}=\mathbf{x}_\mathrm{d}^k \\ \mathbf{u}=\mathbf{u}_\mathrm{d}^k}} ,$$

and the sampling time T_s. The LQR control law

$$\Delta\mathbf{u}^k = \mathbf{K}^k \Delta\mathbf{x}^k , \tag{29}$$

with the time-variant feedback matrix $\mathbf{K}^k$, is calculated by iterating the discrete-time Riccati-equation

$$\mathbf{K}^k = -\left(\mathbf{R}_\mathrm{c} + \left(\mathbf{\Gamma}^k\right)^\mathrm{T}\mathbf{P}^{k+1}\mathbf{\Gamma}^k\right)^{-1}\left(\left(\mathbf{\Gamma}^k\right)^\mathrm{T}\mathbf{P}^{k+1}\mathbf{\Phi}^k\right) \tag{30a}$$
$$\mathbf{P}^k = \left(\mathbf{Q}_\mathrm{c} + \left(\mathbf{\Phi}^k\right)^\mathrm{T}\mathbf{P}^{k+1}\mathbf{\Phi}^k\right) - \left(\left(\mathbf{\Gamma}^k\right)^\mathrm{T}\mathbf{P}^{k+1}\mathbf{\Phi}^k\right)^\mathrm{T}$$
$$\cdot\left(\mathbf{R}_\mathrm{c} + \left(\mathbf{\Gamma}^k\right)^\mathrm{T}\mathbf{P}^{k+1}\mathbf{\Gamma}^k\right)^{-1}\left(\left(\mathbf{\Gamma}^k\right)^\mathrm{T}\mathbf{P}^{k+1}\mathbf{\Phi}^k\right) , \tag{30b}$$

backwards from $k = N-1, \ldots, 1$. The controller is parametrized by the positive definite weighting matrices $\mathbf{Q}_\mathrm{c}$ and $\mathbf{R}_\mathrm{c}$ for the states $\Delta\mathbf{x}$ and the input $\Delta\mathbf{u}$. The matrix $\mathbf{P}^N$ for starting the iteration is found by solving the algebraic Riccati-equation, i. e. (30b) for $k \to \infty$.

6. EXPERIMENT

The experimental implementation for the setup described in Section 2 puts together the results of the previous sections to perform a swing-up and stabilization of the spherical pendulum.

The robot torque input $\mathbf{u}^k$ is adapted to

$$\mathbf{u}^k = \mathbf{u}_\mathrm{d}^k - \mathbf{g}_\mathrm{r}(\mathbf{q}_\mathrm{r}) + \Delta\mathbf{u}^k , \tag{31}$$
$$\mathbf{q}_\mathrm{r} = [\mathbf{E}_7 \; \mathbf{0}_{7\times 11}]\mathbf{x}^k , \tag{32}$$

as the robot controller already compensates the gravitational forces $\mathbf{g}_\mathrm{r}(\mathbf{q}_\mathrm{r})$ based on the joint measurements and the internal robot model when it is interfaced via FRI. Additionally, the robot controller uses a disturbance observer to compensate for most of the friction occurring in the robot joints. Note that the robot and pendulum joint velocities cannot be measured directly, but have to be numerically differentiated using DT_1 filters with a time constant of $T_1 = 15\,\mathrm{ms}$.

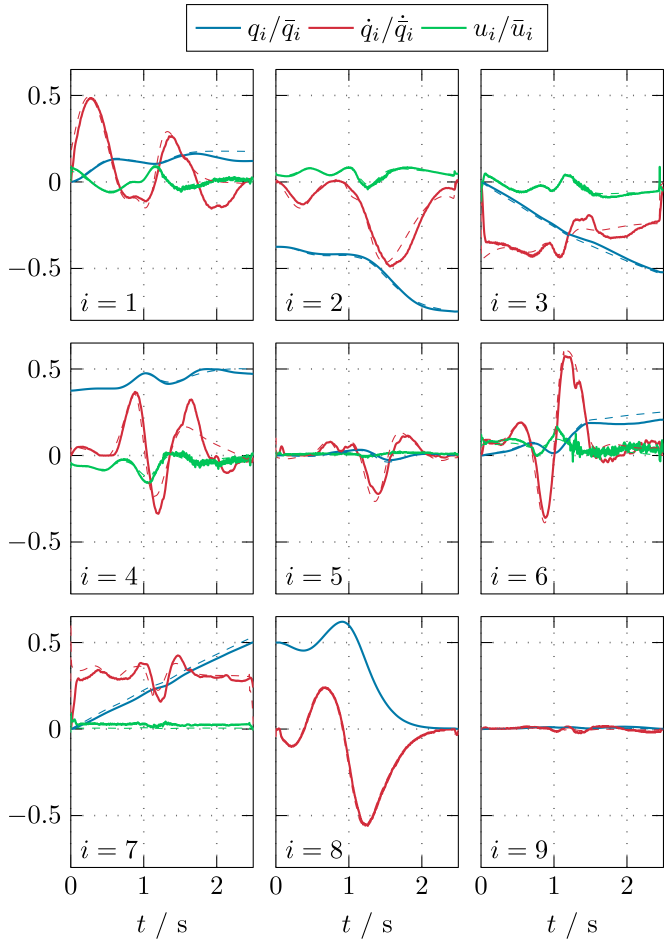

Fig. 5. Experimental results for the swing-up and stabilization of the spherical pendulum. The solid lines display the scaled measurement data and the dashed lines show the desired trajectory from Section 4.3 for $N = 2501$.

Furthermore, the torque input (31) consists of the swing-up trajectory (24), resulting from the optimization problem (23) with $N = 2501$, and the LQR control law (29) with the time-variant feedback matrix $\mathbf{K}^k$ from (30). The LQR is parametrized with the matrices

$$\mathbf{Q}_c = \mathrm{diag}(1, 1, 1, 1, 1, 1, 1, 100, 1, \\ 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 10^{-4}, 10^{-6}) \quad (33a)$$

$$\mathbf{R}_c = 10^{-3} \cdot \mathbf{E}_7 \ , \quad (33b)$$

which were tuned to obtain a good closed-loop performance. The weights for the pendulum velocities $\dot{q}_8$ and $\dot{q}_9$ are reduced to avoid oscillations during swing-up and stabilization. Reducing the weight for the inputs in (33b) allows for larger control inputs and reduces oscillations.

Finally, the experimental results are depicted in Fig. 5 and compared to the desired trajectory from Section 4.3. The graphs are scaled using the corresponding limits from (7). Additionally, a video of the swing-up and stabilization can be found at `www.acin.tuwien.ac.at/9e6f`. While deviations from the desired trajectory are visible, the LQR stabilizes the system around the desired trajectory and achieves a successful swing-up of the spherical pendulum. These control errors are caused by the presence of friction, especially in the pendulum axes, and by the measurement noise in the velocity signals, mainly due to sensor quantization and approximate numerical differentiation.

7. CONCLUSIONS

This paper documents the design of a swing-up control concept for a spherical pendulum using an optimization-based three-step process and demonstrates its successful experimental realization using a custom-built pendulum tool mounted on an industrial 7-axis robot. Future work aims at incorporating time optimality into the controller design, and relaxing kinematic constraints, like the fixed tool frame orientation and the final robot position, as well as exploiting the full range within the velocity and torque limits of the robot.

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